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# Lyapunov-based Stochastic Nonlinear Model Predictive Control: Shaping the State Probability Distribution Functions

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***Abstract*— Stochastic uncertainties in complex systems lead to variability of system states, which can degrade the closed-loop performance. This paper presents a model predictive control approach for a class of nonlinear systems with unbounded stochastic uncertainties. The control approach aims at shaping the probability distribution function of stochastic states, while satisfying input and joint state chance constraints. Closed-loop stability is ensured by designing a stability constraint in terms of a stochastic control Lyapunov function, which explicitly characterizes stability in a probabilistic sense. The Fokker-Planck equation is used for describing the evolution of the probability distribution function of states. Constructing the probability distribution functions using the Fokker-Planck equation allows for shaping the states' distribution functions as well as direct computation of the joint state chance constraints. The closed-loop performance of the stochastic optimal control approach is demonstrated on a benchmark continuous bioreactor.**

## I. Introduction

The need to account for system uncertainties in model predictive control (MPC) of complex dynamical systems has led to extensive investigation of robust MPC approaches (e.g., see [1]). The majority of work on robust MPC considers bounded, deterministic uncertainty descriptions with the goal of designing control laws that are robust to *worst-case* system uncertainties. Deterministic robust MPC approaches may, however, result in conservative closed-loop control performance, as worst-case system uncertainties can have a small probability of occurrence in practice [2]. When probabilistic descriptions of uncertainties are available, stochastic MPC (SMPC) will be a more natural approach to dealing with the stochastic system uncertainties (i.e., parametric uncertainties, uncertain initial conditions, and exogenous disturbances). SMPC approaches allow for incorporating chance constraints into a stochastic optimal control problem to systematically seek tradeoffs between the control performance and robustness to system uncertainties (see [3] for a review on SMPC).

The solution to the stochastic optimal control problem largely depends on the complexity of system dynamics and properties of stochastic uncertainties. SMPC approaches have been proposed for linear systems with multiplicative noise [4], [5] and additive noise [6], [7], [8], [9]. These approaches mostly use affine parameterizations of a feedback control law to transform the stochastic optimal control problem into a deterministic one. Randomized algorithms have also been used to develop SMPC approaches for linear systems [10], [11]. A closed-loop SMPC approach was presented in [12] for nonlinear systems with additive disturbances in the absence of input constraints. Recently, a SMPC approach has been proposed for nonlinear systems with time-invariant probabilistic uncertainties using the generalized polynomial chaos framework [13]. Generally, the characteristics of stochastic uncertainties (e.g., boundedness, additive/multiplicative, and time-varying/time-invariant nature of stochastic uncertainties) have important implications for closed-loop stability and recursive feasibility of SMPC approaches. In addition, the complexity of system dynamics largely affects the computational complexity of probabilistic uncertainty propagation as well as chance constraint handling [3].

This paper presents a stochastic nonlinear MPC (SNMPC) approach for a class of nonlinear systems with probabilistic uncertain initial states as well as unbounded stochastic disturbances. The proposed SNMPC approach includes input constraints and joint state chance constraints. To ensure closed-loop stability, a stochastic Lyapunov-based feedback control law that explicitly characterizes stability in a probabilistic sense is used (e.g., [14], [15]). The Lyapunov-based feedback control law allows for designing a *stability constraint* in terms of a stochastic control Lyapunov function, which guarantees that the origin of the closed-loop system is asymptotically stable in probability (Section II).

The Lyapunov-based SNMPC approach is intended to shape the probability distribution function (PDF) of the stochastic system states. This necessitates constructing the PDF of states. The Fokker-Planck equation [16] is used to describe the dynamic evolution of the (multivariate) PDFs associated with the stochastic nonlinear system (Section III-A). Complete characterization of the PDF of states also allows for direct computation of joint state chance constraints without approximation. This work uses the Hellinger distance [17] to quantify the similarity between the predicted (multivariate) PDF of states and a prespecified reference PDF for shaping the probability distribution functions (Section III-B). Control of the PDF of system states/outputs [18], [19], [20] and MPC of stochastic systems using the Fokker-Planck equation [21], [22] have been reported in the literature. What distinguishes this work is the generic formulation of the Lyapunov-based SNMPC approach in terms of PDF shaping and input and joint state chance constraints handling as well as the ability to ensure closed-loop stability (Section III-C). The presented Lyapunov-based SNMPC approach is demonstrated for stochastic optimal control of a benchmark continuous bioreactor in the presence of stochastic uncertainties (Section IV).

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## II. PRELIMINARIES

### *Notation*

Throughout this paper, boldface symbols (e.g., $\mathbf{x}$) denote vectors and subscripts denote vector elements (e.g., $x_i$). $\mathbb{R}^n$ denotes the $n$-dimensional Euclidean space with $\mathbb{R}_+ = [0, \infty)$. $\mathsf{Tr}\{\cdot\}$ denotes the trace operator on a square matrix. For a vector $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|$ denotes the Euclidean norm of $\mathbf{x}$, and $\|\mathbf{x}\|_{\mathbf{Q}}^2$ denotes the weighted norm of $\mathbf{x}$ defined by $\|\mathbf{x}\|_{\mathbf{Q}}^2 = \mathbf{x}^\top \mathbf{Q}\mathbf{x}$ with $\mathbf{Q}$ being a positive definite symmetric matrix. $\mathbb{E}[\cdot]$ denotes the expectation of a stochastic variable. $\mathsf{P}_{\mathbf{x}}$ denotes the (multivariate) probability distribution function of $\mathbf{x} \in \mathbb{R}^n$. $(\Omega, \mathcal{F}, \mathcal{P})$ denotes a probability space defined by the sample space $\Omega$, $\sigma$-algebra $\mathcal{F}$, and probability measure $\mathcal{P}$ on $\Omega$. $\mathbf{Pr}\{\cdot\}$ denotes the probability of satisfaction of an expression. $\mathcal{L}_{\mathbf{f}}\mathcal{X}$ denotes the Lie derivative of a scalar function $\mathcal{X}(\cdot)$ with respect to a vector function $\mathbf{f}(\cdot)$. A continuous function $\mathbf{V} : \mathbb{R}^n \to \mathbb{R}$ is said to be $C^k$ if it is $k$-times differentiable. A continuous function $\alpha : \mathbb{R}_+ \to \mathbb{R}_+$ is said to belong to class $\mathcal{K}$ when it is strictly increasing and $\alpha(0) = 0$. The function $\alpha$ is said to belong to class $\mathcal{K}_\infty$ when $\alpha \in \mathcal{K}$ and $\alpha(a) \to \infty$ as $a \to \infty$.

### *System description*

Consider a class of stochastic nonlinear systems described by the stochastic differential equation (SDE)

$$\begin{aligned} &d\mathbf{x}(t) = \mathbf{f}(\mathbf{x}(t))dt + \mathbf{g}(\mathbf{x}(t))\mathbf{u}(t)dt + \mathbf{h}(\mathbf{x}(t))d\mathbf{w}(t) \\ &\mathbf{x}(t_0) \sim \mathsf{P}_{\mathbf{x}_0}, \end{aligned} \tag{1}$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ denotes the stochastic state variables with the known initial multivariate PDF $\mathsf{P}_{\mathbf{x}_0}$; $\mathbf{u}(t) \in \mathbb{R}^m$ denotes the system inputs; $\mathbf{w}(t)$ denotes a $q$-dimensional standard Wiener process (i.e., stochastic disturbances) defined on the probability space $(\Omega, \mathcal{F}, \mathcal{P})$; and $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$, $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, and $\mathbf{h} : \mathbb{R}^n \to \mathbb{R}^{n\times q}$ denote the Borel measurable functions that describe the system dynamics. The functions $\mathbf{f}$, $\mathbf{g}$, and $\mathbf{h}$ are assumed to be locally bounded and locally Lipschitz continuous in $\mathbf{x}(t)$, $\forall t \in \mathbb{R}_+$, and $\mathbf{f}(0) = 0$ (i.e., the origin is the steady-state point of the unforced and undisturbed system). The latter conditions ensure uniqueness and local existence of solutions to the SDE (1) [23]. The system inputs $\mathbf{u}(t)$ in (1) are constrained to lie in a nonempty convex set $\mathbb{U} \subseteq \mathbb{R}^m$ defined by

$$\mathbb{U} := \{\mathbf{u}(t) \in \mathbb{R}^m \mid \mathbf{u}_{min} \leq \mathbf{u}(t) \leq \mathbf{u}_{max}\}, \tag{2}$$

where $\mathbf{u}_{min} \in \mathbb{R}^m$ and $\mathbf{u}_{max} \in \mathbb{R}^m$ denote the lower and upper bounds on $\mathbf{u}$, respectively. In addition, the stochastic system states $\mathbf{x}(t)$ should satisfy hard inequality constraints

$$\mathbf{k}(\mathbf{x}(t)) \leq 0, \tag{3}$$

where $\mathbf{k} : \mathbb{R}^n \to \mathbb{R}^p$ denotes (possibly) nonlinear functions that describe the state constraints, and $\mathbf{k}(0) = 0$.

Note that the stochasticity of system (1) arises from the probabilistic nature of uncertain initial states $\mathbf{x}(t_0)$ (described by $\mathsf{P}_{\mathbf{x}_0}$) and the stochastic disturbances $\mathbf{w}$. In (1), the terms $\mathbf{f}(\mathbf{x}(t)) + \mathbf{g}(\mathbf{x}(t))\mathbf{u}(t)$ and $\mathbf{h}(\mathbf{x}(t))$ correspond to the drift and diffusion terms in the *Ito stochastic process*, respectively [24].

### *Stochastic optimal control with state chance constraints*

This paper investigates MPC of the stochastic nonlinear system (1) such that stability of the closed-loop system is guaranteed. The proposed SNMPC approach should allow for shaping the multivariate PDF $\mathsf{P}_{\mathbf{x}}(t)$ of system states in an optimal manner, while the system inputs $\mathbf{u}(t)$ lie in the set $\mathbb{U}$. In addition, the stochastic optimal control approach should ensure satisfaction of the state constraints (3) with at least probability $\beta$ in the presence of system stochasticity. This requires incorporating *joint* chance constraints

$$\mathbf{Pr}\{\mathbf{k}(\mathbf{x}(t)) \leq 0\} \geq \beta \tag{4}$$

into the stochastic optimal control problem. This paper considers receding-horizon implementation of the stochastic optimal control problem in a full state feedback control scheme, where the PDF $\mathsf{P}_{\mathbf{x}}(t_k)$ is available at every measurement sampling time instant $t_k$.

The key challenges that will be addressed in this paper for solving the above described stochastic optimal control problem are: (i) describing the dynamic evolution of the multivariate PDF $\mathsf{P}_{\mathbf{x}}(t)$ associated with the SDE (1), (ii) converting the joint chance constraint (4) to computationally tractable expressions, and (iii) designing the control law such that it ensures closed-loop stability of the stochastic system. Next, the main result of stochastic Lyapunov stability (e.g., see [14], [25]) is summarized. This result is used for designing a Lyapunov-based SNMPC approach.

### *Lyapunov-based controllers*

For stochastic nonlinear systems, Lyapunov-based stabilizing control laws explicitly characterize the region of attraction of the closed-loop system in a probabilistic sense (e.g., see [14], [26], [27], [15], and the references therein). In this paper, a Lyapunov-based control law is designed for the above stochastic optimal control problem. It is assumed that there exists a nonlinear feedback control law $\mathbf{u}(t) = \mathbf{p}(\mathbf{x})$, $\forall \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n$, where $\mathbf{p} : \mathbb{R}^n \to \mathbb{R}^m$ denotes a nonlinear function and $\mathcal{X}$ denotes a compact set that contains the origin $\mathbf{x} = 0$. The feedback control law $\mathbf{p}(\mathbf{x})$ is intended to make the closed-loop system asymptotically stable (in probability) about the origin, while the input and state constraints (i.e., (2) and (4)) are satisfied. According to the converse Lyapunov theorem [28], the existence of the feedback control law $\mathbf{p}(\mathbf{x})$ implies the existence of a *stochastic control Lyapunov function* $\mathbf{V}(\mathbf{x})$ defined as in Thm. 1.

**Theorem 1 (Asymptotic stability in probability [14]):** Consider the stochastic nonlinear system (1) and assume that there exists a $C^2$-function $\mathbf{V} : \mathbb{R}^n \to \mathbb{R}_+$, class $\mathcal{K}_\infty$ functions $\alpha_1$ and $\alpha_2$, and a class $\mathcal{K}$ function $\alpha_3$, such that $\forall \mathbf{x} \in \mathcal{X}$, $\forall t \geq 0$

$$\alpha_1(|\mathbf{x}|) \leq \mathbf{V}(\mathbf{x}) \leq \alpha_2(|\mathbf{x}|),$$

$$\mathcal{L}_{\mathbf{f}}\mathbf{V}(\mathbf{x})+\mathcal{L}_{\mathbf{g}}\mathbf{V}(\mathbf{x})\mathbf{u}(t)|_{\mathbf{u}(t)=\mathbf{p}(\mathbf{x})}+ \frac{1}{2}\mathsf{Tr}\{\mathbf{h}(\mathbf{x})^\top\frac{\partial^2\mathbf{V}}{\partial\mathbf{x}^2}\mathbf{h}(\mathbf{x})\}\le\alpha_3(|\mathbf{x}|).$$

Then, the stochastic control Lyapunov function $\mathbf{V}(\mathbf{x})$ ensures that the origin is asymptotically stable in probability. ■

Thm. 1 indicates that the stochastic Lyapunov-based control technique allows for defining feedback control laws that will lead to

$$\mathcal{L}_{\mathbf{f}}\mathbf{V}(\mathbf{x})+\mathcal{L}_{\mathbf{g}}\mathbf{V}(\mathbf{x})\mathbf{u}(t)|_{\mathbf{u}(t)=\mathbf{p}(\mathbf{x})}+\frac{1}{2}\mathsf{Tr}\{\mathbf{h}(\mathbf{x})^\top\frac{\partial^2\mathbf{V}}{\partial\mathbf{x}^2}\mathbf{h}(\mathbf{x})\} + \gamma\mathbf{V}(\mathbf{x})\le 0, \qquad \forall\mathbf{x}\in\Pi, \tag{5}$$

where the set $\Pi$ is defined by

$$\Pi\coloneqq\sup_{c\in\mathbb{R}}\{\mathbf{x}\in\mathbb{R}^n\mid\mathbf{u}\in\mathbb{U},\ \mathbf{x}\in\mathcal{X},\ \mathbf{V}(\mathbf{x})\le c\};$$

and $\gamma>0$ is a constant. The stochastic control Lyapunov function $\mathbf{V}(\mathbf{x})$ is used for ensuring closed-loop stability of the proposed SNMPC approach.

## III. Stochastic Nonlinear Model Predictive Control

This section presents the formulation of the Lyapunov-based SNMPC approach with joint state chance constraints. The propagation of probabilistic system uncertainties (i.e., uncertain initial states and stochastic disturbances) through system dynamics is performed by the *Fokker-Planck equation*. The *Hellinger distance* is used as a measure of similarity of multivariate PDFs in the objective function of the stochastic optimal control problem to shape the PDF of states.

### A. Fokker-Planck Equation for Uncertainty Propagation

The Fokker-Planck (FP) equation is used to describe the dynamic evolution of the PDF of stochastic states $\mathbf{x}(t)$ in system (1) [16]. The FP equation readily characterizes the complete multivariate PDF $\mathsf{P}_{\mathbf{x}}(t)$ arisen from the stochastic system uncertainties in initial states $\mathbf{x}(t_0)$ (and measured states at every sampling instant) as well as disturbances $\mathbf{w}$. This is in contrast to uncertainty propagation techniques that describe merely certain *statistics* of the PDFs (e.g., see [13] and the references therein). Constructing the PDF of states using the FP equation enables the proposed SNMPC approach to: (i) shape the PDF $\mathsf{P}_{\mathbf{x}}$ with respect to any reference (multivariate) PDF, and (ii) compute chance constraints of any complexity directly without conservative approximations.[1]

The FP equation associated with the SDE (1), which describes the evolution of the multivariate PDF $\mathsf{P}_{\mathbf{x}}(t)$, is defined by

$$\frac{\partial\mathsf{P}_{\mathbf{x}}}{\partial t}+\sum_{i=1}^{n}\frac{\partial}{\partial x_i}\Big(\big(f_i(\mathbf{x})+g_i(\mathbf{x})\mathbf{u}\big)\mathsf{P}_{\mathbf{x}}\Big) -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2}{\partial x_i\partial x_j}\Big(D_{ij}(\mathbf{x})\mathsf{P}_{\mathbf{x}}\Big)=0 \tag{6}$$

[1]The only numerical approximation will be associated with discretization of the FP equation.

with the initial condition

$$\mathsf{P}_{\mathbf{x}}(0)=\mathsf{P}_{\mathbf{x}_0},$$

where $\mathbf{D}=\mathbf{h}(\mathbf{x})\mathbf{h}(\mathbf{x})^\top$ denotes the diffusion matrix. The FP equation (6) is a parabolic partial differential equation, whose solution should be nonnegative and satisfy

$$\int_{\Omega_{\mathbf{x}}}\mathsf{P}_{\mathbf{x}}(t)d\mathbf{x}=1,\quad\forall t\ge 0.$$

The existence and uniqueness of a solution to (6) under mild assumptions have been established [16], [29]. The FP equation (6) can be used to compute the univariate PDFs $\mathsf{P}_{x_i}$ for every state $x_i$ as well as joint PDFs for any combination of stochastic states.

Solving the FP equation is generally challenging for nonlinear systems, in particular systems with high state dimension [16]. Various numerical methods such as finite difference and finite element methods have been used to solve the FP equation for nonlinear systems (e.g., see [30] and the references therein). In this work, the finite volume method with first order upwind interpolation scheme [31] is used to solve (6). The finite volume method allows for effectively dealing with the convective nature of the FP equation (due to the drift term) and suppressing numerical instability problems.

### B. Metric for Similarity of Probability Distribution Functions

The proposed SNMPC approach aims at shaping the PDF $\mathsf{P}_{\mathbf{x}}$ according to a reference (arbitrarily-shaped) PDF. Hence, a metric is required to quantify the similarity between the predicted and the reference PDFs at each time instant $t$. This paper uses the Bhattacharyya coefficient [32], a measure closely related to the *Bayes error* [33], to establish a metric for the similarity of PDFs. The Bhattacharyya coefficient, which quantifies the degree of overlap between two (multivariate) PDFs, is defined by

$$\mathfrak{B}(\mathbf{x})\coloneqq\int_{\Omega_{\mathbf{x}}}\sqrt{\mathsf{P}_{\mathbf{x}}(t)\mathsf{P}_{\mathbf{x}}^{\mathbf{ref}}}d\mathbf{x}, \tag{7}$$

where $\mathsf{P}_{\mathbf{x}}^{\mathbf{ref}}$ denotes the reference PDF. The Bhattacharyya coefficient will be larger when the overlap between the PDFs is larger. $\mathfrak{B}(\mathbf{x})=0$ if the PDFs do not overlap, whereas $\mathfrak{B}(\mathbf{x})=1$ when the PDFs are identical. Explicit forms of the Bhattacharyya coefficient for various PDFs are given in [32], [34].

The Bhattacharyya coefficient (7) is used to define a metric for quantifying the similarity between $\mathsf{P}_{\mathbf{x}}(t)$ and $\mathsf{P}_{\mathbf{x}}^{\mathbf{ref}}$ in the objective function of the stochastic optimal control problem. The metric, known as the Hellinger distance [17], is defined by

$$\Delta(\mathbf{x})\coloneqq\sqrt{1-\mathfrak{B}(\mathbf{x})}. \tag{8}$$

The metric (8) is near optimal due to its relation to the Bayes error, and can be used for arbitrary PDFs [35].

### C. Formulation of the Lyapunov-based SNMPC Approach

The Fokker-Planck equation (6) and the Hellinger distance (8) are used to cast the Lyapunov-based stochastic optimal control problem for the stochastic nonlinear system (1).

**Problem 1 (Lyapunov-based SNMPC with input and joint state chance constraints):** Suppose that the PDF $\mathsf{P}_{\mathbf{x}}(t_k)$ is known at every sampling time instant $t_k$.[2] The stochastic optimal control problem at each time instant $t_k$ is stated as

$$
\begin{aligned}
&\mathbf{u}^*(\mathsf{P}_{\mathbf{x}}(t_k)) := \arg\min_{\mathbf{u}} \int_0^{T_p} \Big(\Delta(\bar{\mathbf{x}}(\tilde{t})) + \big\|\mathbf{u}(\tilde{t})\big\|_{\mathbf{R}}^2\Big) d\tilde{t} \\
&\text{s.t.:} \quad \frac{\partial \mathsf{P}_{\bar{\mathbf{x}}}}{\partial t} + \sum_{i=1}^{n} \frac{\partial}{\partial \bar{x}_i}\Big(\big(f_i(\bar{\mathbf{x}}) + g_i(\bar{\mathbf{x}})\mathbf{u}\big)\mathsf{P}_{\bar{\mathbf{x}}}\Big) \\
&\qquad -\tfrac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial^2}{\partial \bar{x}_i \partial \bar{x}_j}\Big(D_{ij}(\bar{\mathbf{x}})\mathsf{P}_{\bar{\mathbf{x}}}\Big) = 0, \qquad \forall t \in [0,\ T_p] \\
&\qquad \mathcal{L}_{\mathbf{f}}\mathbf{V}(\bar{\mathbf{x}}) + \mathcal{L}_{\mathbf{g}}\mathbf{V}(\bar{\mathbf{x}})\mathbf{u}(t) + \\
&\qquad \tfrac{1}{2}\mathsf{Tr}\{\mathbf{h}(\bar{\mathbf{x}})^\top \tfrac{\partial^2 \mathbf{V}}{\partial \bar{\mathbf{x}}^2}\mathbf{h}(\bar{\mathbf{x}})\} + \gamma\mathbf{V}(\bar{\mathbf{x}}) \leq 0, \qquad \forall t \in [0,\ T_p] \\
&\qquad \mathbf{Pr}\{\mathbf{k}(\bar{\mathbf{x}}(t)) \leq 0\} \geq \beta, \qquad \forall t \in [0,\ T_p] \\
&\qquad \mathbf{u}(t) \in \mathbb{U}, \qquad \forall t \in [0,\ T_c] \\
&\qquad \mathsf{P}_{\bar{\mathbf{x}}}(0) = \mathsf{P}_{\mathbf{x}}(t_k),
\end{aligned}
\tag{9}
$$

where $\mathbf{u}^*$ denotes the vector of optimal inputs (i.e., control policy) over the control horizon $[0, T_c]$; $\bar{\mathbf{x}}$ denotes the stochastic states predicted by the system model; $T_p$ denotes the prediction horizon; and $\mathbf{R}$ denotes a strictly positive definite matrix. The closed-loop stability of the SNMPC approach is ensured in a probabilistic sense by incorporating the stability constraint (defined in terms of the stochastic control Lyapunov function $\mathbf{V}(\bar{\mathbf{x}})$) into (9). The asymptotic stability properties of the closed-loop system are characterized in Thm. 1 and (5). The proposed stochastic control approach possesses the stability properties of Lyapunov-based controllers when applied in a sample-and-hold fashion (e.g., see [15]). The stochastic optimal control problem (9) is implemented in a receding-horizon mode, where merely the optimal inputs $\mathbf{u}^*(0)$ are applied to the stochastic nonlinear system (1) at every sampling time instant $t_k$.

The Lyapunov-based SNMPC approach in Problem 1 enables shaping the multivariate PDF of states, while satisfying input constraints and joint state chance constraints. This is due to using the FP equation for probabilistic uncertainty propagation since the FP equation allows for constructing the PDF of the states. The objective function in (9) is stated in terms of the Hellinger distance to quantify the difference between the multivariate PDF $\mathsf{P}_{\mathbf{x}}(t)$ and the reference distribution function $\mathsf{P}_{\mathbf{x}}^{\mathbf{ref}}$ over the prediction horizon $t \in [0,\ T_p]$.[3] The objective function can be simplified by considering only certain statistics (e.g., the expected value and variance) of the PDFs. Furthermore, the joint state chance constraint in (9) can be readily computed through explicit integration without conservative approximations since the knowledge of the full multivariate PDF is available. When only individual state chance constraints are considered in (9), the FP equation can be adapted to compute only the univariate PDFs for the respective states.

[2]Notice that this work considers a full state feedback control scheme. The PDFs $\mathsf{P}_{\mathbf{x}}(t_k)$ arise from measurement error.

## IV. Case Study: Stochastic Optimal Control of a Continuous Bioreactor

The performance of the proposed SNMPC approach is demonstrated on a benchmark continuous bioreactor [36], [37]. The dynamics of the bioreactor are described by

$$
\begin{aligned}
dX &= \Big(-\frac{F_{in}}{V} + \mu X\Big)dt \\
&\quad + \sigma_X dw_X(t), \quad X(0) \sim \mathcal{N}(\bar{X}, 0.1) \\
dS &= \Big(\frac{F_{in}}{V}(S_f - S) - \frac{1}{Y_{X|S}}\Big)dt \\
&\quad + \sigma_S dw_S(t), \quad S(0) \sim \mathcal{N}(\bar{S}, 0.1) \\
dP &= \Big(-\frac{F_{in}}{V}P + (\alpha\mu + \beta)X\Big)dt \\
&\quad + \sigma_P dw_P(t), \quad P(0) \sim \mathcal{N}(\bar{P}, 0.1) \\
dh &= \Big(\frac{1}{\pi r^2}(F_{in} - k_1\sqrt{h})\Big)dt, \quad h(0) = \bar{h},
\end{aligned}
$$

where $X$ is the biomass concentration; $S$ is the substrate concentration; $P$ is the product concentration; $h$ is the level of the reactor; $F_{in}$ is the flow rate of the substrate into the reactor; and $S_f$ is the concentration of the substrate in the feed stream. $w_X, w_S,$ and $w_P$ denote standard Wiener processes operating on the system. The kinetic parameter $\mu$ and the reactor volume $V$ are defined by

$$
\mu = \frac{\mu_0(1 - \frac{P}{P_m}S)}{K_m + S + \frac{S^2}{K_i}} \text{ and } V = \pi r^2 h,
$$

respectively, where $\mu_0$ is the maximum specific growth rate of the biomass in the reactor. The model parameters are listed in Table I.

The bioreactor under study is a stochastic system due to the additive white process noise $w_i \sim \mathcal{N}(0, 0.02), i = \{X, S, P\}$ and the uncertain initial conditions $X(0)$, $S(0)$, and $P(0)$. It is assumed that the concentrations of $X, S,$ and $P$ as well as the reactor level $h$ are measured at every sampling time instant $t_k$, that is, every 1 hr. The measured outputs are corrupted by zero-mean measurement

[3]When the control objective is defined in terms of PDF shaping for individual states, the objective function in (9) must be modified to weighted sum of Hellinger distance defined for each state.

TABLE I

BIOREACTOR MODEL PARAMETERS.

| | | | |
|---|---|---|---|
| $Y_{X\|S}$ | 0.4 g/g | $K_i$ | 22 g/L |
| $\alpha$ | 2.2 g/g | $F_{in}$ | 0.164 L/min |
| $\beta$ | 0.2 hr$^{-1}$ | $\bar{X}$ | 7.03 g/L |
| $\mu_0$ | 0.48 hr$^{-1}$ | $\bar{S}$ | 6.03 g/L |
| $P_m$ | 50 g/L | $\bar{P}$ | 23.3 g/L |
| $K_m$ | 1.2 g/L | $\bar{h}$ | 1.18 m |
| $\sigma_X$ | 0.02 g/L | $\sigma_S$ | 0.02 g/L |
| $\sigma_P$ | 0.02 g/L | | |

noise with normal distribution $\mathcal{N}(0, 0.1)$. The only system input that can be manipulated for control is $S_f$. The process is run for 70 hr.

The control objective is to shape the PDF of the product concentration around the reference distribution $\mathsf{P}_P^{ref} = \mathcal{N}(\bar{P}, \Sigma_P)$, where $\bar{P} = 23.3$ g/L and $\Sigma_P = 0.1$. The objective function is defined as

$$\int_0^{T_p} \left( \Delta(\mathsf{P}(P(\tilde{t})), \mathsf{P}_P^{ref}) + \left(\mathbb{E}[P(\tilde{t})] - P_{mes}(\tilde{t})\right)^2 \right) d\tilde{t}, \tag{10}$$

where $P_{mes}$ denotes the measured product concentration. The objective function (10) is defined such that the SNMPC approach minimizes the difference between the PDF of $P$ and its reference PDF $\mathsf{P}_P^{\mathbf{ref}}$ over the prediction horizon $[0,\ T_p]$. An integrating action is conferred to the controller by including the term $\left(\mathbb{E}[P(\tilde{t})] - P_{mes}(\tilde{t})\right)^2$ into the objective function (10) [38]. The prediction horizon and the control horizon are selected to be $T_p = 20$ hr and $T_c = 20$ hr, respectively. To design the stability constraint, the quadratic Lyapunov function $\mathbf{V}(\mathbf{x})$ is defined as $\mathbf{V}(\bar{\mathbf{x}}) = \bar{\mathbf{x}}^\top \mathbf{P} \bar{\mathbf{x}}$, where $\bar{\mathbf{x}} = [X - \bar{X} \;\; S - \bar{S} \;\; P - \bar{P} \;\; h - \bar{h}]^\top$ and

$$\mathbf{P} = 10^3 \times \begin{bmatrix} -0.0899 & -0.0060 & 0.0215 & -3.3081 \\ -0.0060 & 0.0032 & 0.0009 & 0.1017 \\ 0.0215 & 0.0009 & -0.0014 & 0.4369 \\ -3.3081 & 0.1017 & 0.4369 & 7.6834 \end{bmatrix}.$$

The matrix $\mathbf{P}$ is obtained by solving the Lyapunov equation for the (nominal) linearized system dynamics around the steady-state point. The parameter $\gamma$ in the stability constraint is chosen to be $10^8$.

In the stochastic optimal control problem, the input to the system is constrained to lie in the range 15 g/L $\leq S_f \leq$ 35 g/L. The lower bound on $S_f$ is applied to avoid starvation of the biomass. In addition, the concentration of substrate $S$ should remain below an admissible threshold in the presence of probabilistic system uncertainties. Hence, the chance constraint

$$\mathbf{Pr}\{S(t) \geq 10.5 \text{ g/L}\} \leq 0.05$$

is incorporated, implying that the state constraint $S(t) \leq$ 10.5 g/L should be satisfied with at least probability 95% in a stochastic setting. The FP equation is solved using the finite volume method with 200 discretization points over the concentration supports $X_{sup} = [5$ g/L, $8$ g/L$]$, $S_{sup} = [0$ g/L, $15$ g/L$]$, and $P_{sup} = [10$ g/L, $30$ g/L$]$. The above constrained nonlinear optimal control problem is solved using the MATLAB subroutine `fmincon`, where the set of model equations is integrated using the solver `ODE15s`. The control input is discretized in a piecewise constant fashion in 4 intervals over the control horizon.

To evaluate the performance of the SNMPC approach, a closed-loop simulation run is performed. Fig. 1 shows the evolution of the probability distribution of the product concentration $P$ in the course of the process. The SNMPC approach stabilizes the stochastic system around the steady-state operating point ($\bar{P}$ in Fig. 1). The integrating action of the controller reduces the offset compared to the case of only using the Hellinger distance as the objective function (not shown). Fig. 1 shows that the SNMPC approach can effectively shape the PDF of $P$ according to the reference PDF $\mathsf{P}_P^{\mathbf{ref}}$. At the final process time, the mean and variance of the PDF for $P$ are 23.4 and 0.1, respectively, which are in agreement with those of the reference PDF $\mathcal{N}(23.3, 0.1)$. The time evolution of the PDF of substrate concentration is shown in Fig. 2. The substrate concentration remains less than its constraint at all times in the course of the process. Effective constraint handling is due to the state chance constraint, which ensures constraint satisfaction with a desired probability level in the presence of stochastic uncertainties.

## V. CONCLUSIONS

A model predictive control approach is presented for a class of nonlinear systems with stochastic uncertainties. The closed-loop stability of the control approach is ensured by explicitly characterizing stability in a probabilistic sense using a stochastic control Lyapunov function. A key challenge in SMPC is efficient propagation of uncertainties through the system dynamics to construct the probability distribution of the stochastic states. This paper uses the Fokker-Planck equation for uncertainty propagation, which allows for describing the evolution of the probability distribution of the stochastic states for general probabilistic uncertainty descriptions. This work demonstrates that constructing the complete probability distribution enables shaping the probability distribution of

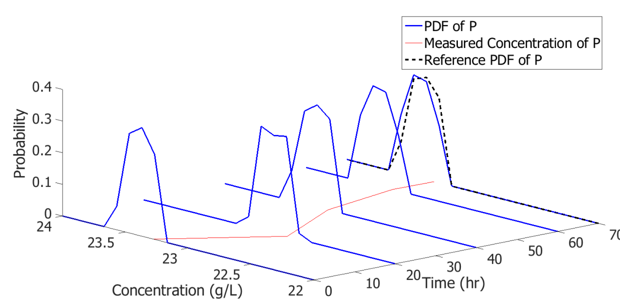


Fig. 1. Probability distribution functions of the product concentration $P$ at various times in the closed-loop simulation run. The SNMPC approach aims at shaping the probability distribution of $P$ according to the reference probability distribution (black dashed distribution).

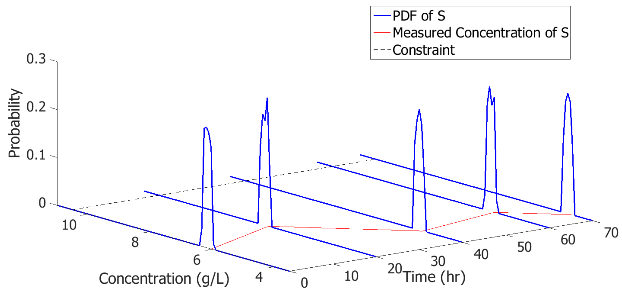


Fig. 2. Probability distribution functions of the substrate concentration $S$ at various times in the closed-loop simulation run. The substrate concentration remains less than the constraint $10.5$ g/L at all times.

states with respect to arbitrarily-shaped reference distributions, as well as direct computation of chance constraints without conservative approximations.

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